



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CCS339 Cryptocurrency and Blockchain Technologies

Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit III / Proof of Stake, Proof of Burn, Proof of Elapsed Time

Course Outcome: CO 3

Unit Learning Outcome: UO 3a

Activity Chosen: Think Pair Share

Justification:

- Think-pair-share is a collaborative learning strategy where students work together to share their understanding of a particular topic that was taught on the day of the class. This strategy requires students to (1) think individually about a topic and (2) share ideas with classmates. Discussing with a partner maximizes participation, focuses attention, and engages students in comprehending the reading material. The students' individual understanding capability can be tested, and the ideas gained can be shared through a think-pair-share activity.
- Time Allotted for the Activity:** 15 minutes

Details of the Implementation:

- The instructor explained the particular concepts or topic in the classroom within 30 minutes. The instructor asked the students to think about the topic they had learned.
- Students think about what they know or have learned in Proof of Stake, Proof of Burn, and Proof of Elapsed Time and come up with their own ideas. [Takes 1–5 minutes].
- The instructor asks them to pair with nearby students to discuss the topic. The students interact with each other about their programs to deepen their understanding of the consensus algorithm.
- They share their thoughts with each other and proceed with the task. [Takes 6–10 minutes].
- Students share their knowledge with the entire class [it takes 11–15 minutes].
- The instructor moderates the discussion and highlights important points.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO9	PSO1
CO 3	2	1	1	1	2	1

(1 – Low 2 – Moderate 3 – High)
PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO9	PSO1
	2	1	1	1	2	1
Justification for correlation	Students will be able to understand the concept of Bitcoin Consensus aligning with Engineering Knowledge	Students will be able to explain the fundamentals consensus mechanism with the help of problem analysis.	Students will gain skill in deploying, and interacting with Consensus algorithms.	Students will apply research-based knowledge and methodologies to validate the effectiveness and performance of various consensus mechanisms.	Students will collaborate in teams to find out the innovative solutions with consensus algorithms	Students will continuously engage in lifelong learning and stay updated with the latest advancements in blockchain consensus research.

• **Images / Screenshot of the practice:**



Fig:1 Glimpse of Think Pair Share Activity

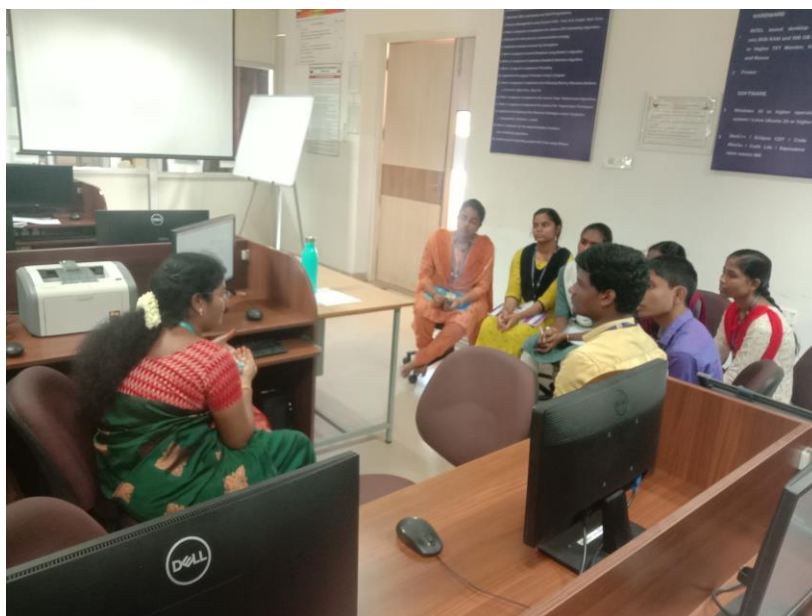


Fig:2 Highlights important points by Instructor

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- M.Mahesh Selvalakshmi shared that the Think-Pair-Share activity gives her the opportunity to feel more comfortable sharing their thoughts.
 - S.Thangaselvi expressed that it improves speaking and listening skills.

- ❖ **Benefit of the practice:**

- Through this activity, students will be able to clarify certain concepts and get greater understanding of the subject matter.
 - Compared to the usual approach, this technique has increased student interaction in the classroom.

- ❖ **Challenges faced in implementation:**

- The activity requires additional time for students to think, pair, and share.

- ❖ **References:**

- 1. <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>
 - 2. <https://www.readingrockets.org/classroom/classroom-strategies/think-pair-share>
 - 3. <https://www.ritrjpm.ac.in/images/computer-science/TPS-Venkat-GE6075.pdf>
 - 4. <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>

Signature of Faculty Member

HOD